Sudheer Kumar Papineni
Sr. Full stack Developer
papinenisudheerkumar@gmail.com
LinkedIn id: www.linkedin.com/in/papinenisudheerkumar

Summary

- Versatile and results-driven **Full Stack / Frontend Developer** with 15+ years of experience building enterprise-grade SaaS applications across Finance & Investments, Payments, Pharma & Healthcare domains.
- Skilled in developing **scalable and high-performance web applications** using **React.js**, **TypeScript**, **Next.js, Node.js (Express)**, **GraphQL**, **HTML5**, **CSS3**, **JavaScript (ES6+)**, and **Bootstrap**.
- Adept at driving the complete Software Development Life Cycle (SDLC) using **Agile/Scrum**, and delivering solutions aligned with **CI/CD** and **DevOps** principles.
- Proven ability to translate complex business workflows into intuitive, low-code/no-code style user interfaces, ensuring clean **UI/UX**, modular code, and **performance optimization**.
- Hands on experience on building high scalable modern applications using **Next JS**, for optimization, client and server rendering, advanced routing.
- Experienced in building **Single Page Applications (SPAs)** with reusable, modular components, **Redux Toolkit**, and **React Hooks**, focused on accessibility and responsive design.
- Expertise in creating **micro-frontend applications** using react & typescript.
- Implemented & created node packages to use internally with high scalable features.
- Strong expertise in **modern JavaScript**, including **ES6**+, **DOM** manipulation, asynchronous programming, and advanced array/object operations.
- Expertise in developing scalable, responsive, and maintainable enterprise user interfaces using **Tailwind CSS**, **Material UI (MUI)**, **SASS/SCSS**, and **LESS**.
- Experienced in building reusable component libraries, implementing design systems, theme-based architectures, and ensuring WCAG-compliant accessibility standards.
- Proficient in integrating **REST** and **GraphQL APIs**, optimizing data flow between client and server, and improving performance and responsiveness.
- Implemented **interactive dashboards and visualizations** using **D3.js** and **Chart.js** for data-driven insights and analytics.
- Implemented flow designs using **React Flows**.
- Hands on experience with **Java Springboot** & **Python** (API side), for development & debugging.
- Migrated large-scale React projects from **JavaScript to TypeScript**, improving reliability, maintainability, and type safety.
- Leveraged **AI-powered tools** such as **GitHub Copilot**, **Google Gemini**, and **ChatGPT** to accelerate code generation, improve debugging, and enhance developer productivity.
- Hands-on experience with **automated testing** (Jest, React Testing Library, Cypress), **cloud deployments** (Azure/AWS), and **Git-based workflows** with **GitHub** and **GitLab**.
- Developed **Node.js middleware and GraphQL APIs** using **PostGraphile**, seamlessly integrating with **PostgreSQL** to enable secure, high-performance data access and schema-**driven automation**.
- Strong communicator with proven ability to collaborate cross-functionally, mentor peers, and deliver accessible, high-performance, and user-centric interfaces.

Education

Masters in Computer Applications, Osmania University, India, 2006.

Bachelors in Science from S.V University, India, 2003.

Learnings & Certifications

Please verify **www.linkedin.com/in/papinenisudheerkumar**

GitHub

https://github.com/sudir81

Technical Skill Sets

Languages	JavaScript (ES6+), TypeScript, Node.js, GraphQL
Front-End	React.js, Next.js , Redux Toolkit, HTML5, CSS3, tailwind, Bootstrap, Material UI, D3.js, Chart.js,
Back-End	Node.js (Express), RESTful APIs, GraphQL APIs, Java, Springboot, basic python
API Integration	REST, GraphQL, Axios, Fetch, Redux-Thunk, RTK Query
Build & Tools	RSBuild, Webpack, Vite, npm, yarn, Babel, ESLint, Prettier
Testing	PlayWright, React Testing Library, Mocha, Cypress
Project Management Tools	JIRA, Confluence, SharePoint
Databases	SQL Server, Postgres SQL
DevOps & Cloud	Azure (App Services, Blob Storage, DevOps Pipelines), Docker, Kubernetes
Version Control	Git, GitHub, GitLab, SourceTree, SVN
IDEs & Utilities	Visual Studio Code, Postman, Chrome DevTools

Professional Experience

US.Bank, Irving, TX **09/2024 – Till now**

Lead developer

Project: ***CDA-UI Self-Service*** *- An internal low-code platform enabling users to onboard data and configure end-to-end data pipelines with minimal manual intervention.*

Project: Airflow Dags Automation

Responsibilities:

- Designed, developed, and enhanced scalable enterprise applications using **React.js, TypeScript, Redux Toolkit, Node.js, and microservices architecture**, delivering modular, maintainable, and high-performance solutions.
- Designed and implemented **micro-frontend architectures using Webpack Module Federation**, enabling modular UI development, reusable components, and independent application integration across enterprise products.

- Established reusable UI components and consistent frontend development patterns using **React, TypeScript, Tailwind CSS, Material UI (MUI), SASS/SCSS, and LESS**, improving maintainability, design consistency, and development efficiency.
- Architected and integrated **RESTful and GraphQL APIs** to support efficient communication between frontend applications and backend services, following API design, validation, error-handling, and performance best practices.
- Designed and developed backend services using **Node.js and Express.js**, implementing middleware, request validation, authentication, centralized exception handling, and reusable service components.
- Developed and maintained **Java/Spring Boot microservices** using controller, service, and repository layers, promoting separation of concerns, maintainable business logic, and consistent REST API design.
- Implemented API security using **JWT-based authentication and role-based access control (RBAC)**, enforcing authorization rules and securing communication between frontend applications and backend services.
- Automated Apache Airflow DAG registration, deployment, and management through a micro-frontend application, streamlining workflow orchestration and reducing manual operational effort.
- Collaborated across frontend and backend services, including Spring Boot APIs and Python services, to troubleshoot integration issues, validate data flows, resolve defects, and improve application reliability.
- Optimized frontend performance through lazy loading, route-based code splitting, React.Memo, memoization, and bundle optimization, contributing to a **30% reduction in page load times**.
- Implemented accessibility and responsive design best practices using semantic HTML, ARIA attributes, keyboard navigation, and accessible UI components, aligning interfaces with applicable WCAG 2.1 requirements.
- Established and maintained unit, integration, and end-to-end testing practices using **Jest, React Testing Library, Playwright, and Cypress**, strengthening regression coverage and application quality.
- Contributed to CI/CD modernization by migrating pipelines from Jenkins to **GitLab CI/CD** and integrating automated build, testing, and deployment workflows with Docker, Kubernetes, and Azure Cloud.
- Applied engineering standards and code-quality practices using **Git, ESLint, Prettier,** and **SonarQube**, conducting peer reviews and promoting readable, maintainable, and reliable code.
- Collaborated with UX designers, product managers, architects, and engineering teams to translate business requirements and complex analytics workflows into scalable technical solutions.
- Participated in architecture and design discussions, technical reviews, sprint planning, and Agile/Scrum ceremonies, supporting delivery planning, technical alignment, and continuous improvement.

T.Rowe Price/Tek Systems, Baltimore, MD 07/2020 – 08/2024
Lead developer

Responsibilities:

- Developed **responsive SPAs** using **React.js, Next.js** , **Redux**, and **GraphQL**, integrating with APIs for real-time data rendering.
- Created and maintained **SASS/SCSS** and **LESS** stylesheets to implement modular, scalable, and theme-driven UI designs.
- Built **dynamic dashboards, data grids, charts, and filters** using **D3.js** and **Chart.js**, enabling visual exploration of customer engagement patterns.

- Designed and developed **RESTful** API services using **Node.js** and **Express.js**, implementing middleware, request validation, and error-handling layers to support scalable backend functionality.
- Implemented advanced UI behaviors — form validation, animations, and reusable widgets — to enhance user experience and interactivity.
- Improved application performance through **lazy loading**, **virtual scrolling**, and **API request batching** for high-volume datasets.
- Developed and maintained **automated test suites** using **Jest** and **React Testing Library**, ensuring > 90 % component coverage and cross-browser reliability.
- Partnered with UX designers and product teams to transform data-heavy analytics into actionable, user-friendly interfaces.
- Enhanced build processes with **Webpack**, **npm/yarn**, and optimized asset pipelines to reduce bundle size and load time.
- Contributed to design discussions, sprint planning, and peer reviews in an **Agile**/**Scrum** environment, ensuring timely, high-quality delivery.

Change Healthcare/Innova Solutions, KOP, PA 02/2020 – 06/2020
Sr. Frontend developer

Responsibilities:

- Lead the team by providing solutions & ideas.
- Design, develop and test HTML5, CSS3, Bootstrap, JavaScript and React.JS that meets accessibility and browser standards for web applications.
- Developed SPA (single page application) by using React JS, JSX, bootstrap.
- Designed & integrated Redux state management library with Redux-thunk as middleware to solve data storage & sharing across components in application.
- Used Apollo client to interact with GraphQL server, and integrated GraphQL queries & schemas to fetch the data to UI.
- Integrated and used Axios & fetch for RESTful API communication.
- Implemented react JS code which handle cross browser compatibility issues in IE, Safari and Chrome.
- Used Jest & Enzyme for Unit testing the application.
- Involved in requirements discussion with managers & product owners.
- Reviewing the design documents, wireframes and PBI's (TFS) stories.
- Extensively used Bitbucket for code commit, build & release process.
- Used debugging tools such as Chrome Dev tools, React developer tools, Redux developer tools & logger to investigate the issues and fix the bugs.
- Interacted with Testing Team, Scrum Masters and Business Analysts for fixing of Issues.

Clean Harbors Inc., Norwell, MA 11/2018 – 01/2020
Lead UI & API (c#.net) developer
Responsibilities:

- Lead the team by providing solutions & ideas.
- Design, develop and test HTML5, CSS3, Bootstrap, JavaScript and React.JS that meets accessibility

and browser standards for web applications.

- Developer UI by using React JS, Flux for developing Single Page Applications (SPA).
- Used React-Router to turn application into Single Page Application
- Implemented api's using C#.net & hands on experience in RabbitMQ to handle queuing systems.
- Handling build & deploy process of application in AWS.
- Implemented react JS code which handle cross browser compatibility issues in IE, Safari and Chrome.
- Worked in using React JS components, Forms, Events, Keys, Router, Animations and Flux concept.
- Integrated and used Axios & react fetch for RESTful API communication.
- Used Jest & Enzyme for Unit testing the application.
- Integrated and worked on React Redux/ Redux Thunk for state management in the application.
- Involved in requirements discussion with managers & product owners.
- Preparing and completing action plans; providing solutions for the requirements, identifying and analyzing problems.
- Reviewing the design documents, wireframes and PBI's (TFS) stories.
- Extensively used TFS & GIT for code commit, build & release process.
- Monitoring the offshore work and providing technical help to the offshore team using TFS backlogs.
- Interacted with Testing Team, Scrum Masters and Business Analysts for fixing of Issues.

Johnson & Johnson, Raritan, NJ **08/2017 – 10/2018**

Lead – Mobile & UI developer
Asset tracking Dashboard application (Web application)

Responsibilities:

- Involved in requirements discussion with managers & product owners.
- Preparing and completing action plans; providing solutions for the requirements, identifying and analyzing problems.
- Developer UI by using React JS for developing Single Page Applications (SPA).
- Used React-Router to turn application into Single Page Application
- Worked in using React JS components, Forms, Events, Keys, Router, Animations and Flux concept.
- Integrated Axios & react fetch for RESTful API communication.
- Integrated and worked on React Redux/ Redux Thunk for state management in the application.
- Reviewing the design documents, wireframes and JIRA stories.
- Identifying & prioritizing new Web features and applications in conjunction with business leaders and department managers
- Involved in the development of RESTful services using Node.js.
- Created cucumber test scripts for CICD process.
- Involved in the creation of JIRA stories, Bitbucket repositories for CICD process.
- Used D3 library for graphs creation & earthjs library for globe implementation for web app.
- Integrated Slack into application, through webhooks & web API.
- Involved in the process of build & release process.

Western Union, Montvale, NJ **09/2015 – 07/2017**

Lead - Desktop & mobile web and Hybrid application
Link: https://iwgo.westernunion.com

Responsibilities:

- Involved in requirements discussion with Business analysts & product owners.
- Preparing and completing action plans; providing solutions for the requirements, identifying and analyzing problems.
- Reviewing the design documents, USX, code review, cross verifying the functionality.
- Identifying & prioritizing new Web features and applications in conjunction with business leaders and department managers
- Proposed new detailed oriented solutions functional/ non-functional, and ideas based on the requirement & current trend in the collaboration with the team.
- Involved in some development part of hybrid applications to support iOS & android platforms.
- Involved in finding & fixing memory leaks by using firebug & developer tools.
- Used Test Driven Development techniques to provide reliable, error free, working software early and often.
- Involved in the design of offline support with cache memory like Indexed DB and integrated geolocation (online & offline).
- Coordination with QA & UAT team for the new features & requirements discussion.
- Daily touch base with project stakeholders, business team & development team to ensure quality of the product is met.
- Involved in the application testing using various mobile technologies including iPhone and Android.
- Involved in the submission of hybrid application in app store & play store, also involved in firebase analytics & iTunesConnect for review the iOS application.

Western Union Digital, San Francisco, CA 09/2011 – 08/2015

Senior Software Engineer

Responsibilities:

- Involved in the understanding of requirement specification in analyzing Business Requirements, System Requirement documents, blue prints for developing Test Cases.
- Involved in the flow design of the application along with client.
- Involved in the requirements and scope discussions with the team.
- Result of drastic depart from traditional web design using REST web services.
- Implementing iOS app which post data to RESTFul web-service implemented in java jersey.
- Worked with Corelocation and MapKit framework for showing agents in current locations on a map on iPhone applications.
- Implement NSURLConnection to integrate with backend web services & used SQLite for data storage.
- Involved in the device detection with Iovation framework to identify fraud transactions.
- Involved in the implementation of ID scan with Jumio framework.

- Involved in push notification services for new versions & features.
- Involved in performance testing and Functional tests.
- Testing included simulator and device testing with iPhone, iPad, and iPod Touch with iOS 8 and later.
- Involved in the development of re-factoring from Objective C to Swift code.
- Involved in the implementation of apple push notification services (APNS) in the app to get the app version updates.
- Used Storyboards and autolayout in the UI development.
- Assisting the QA team with new product releases and bug fixes for products.
- Being an onsite coordinator, worked with offshore team to achieve the targets.
- Expertise of testing the application developed using various mobile technologies including iPhone and Android.
- Perform root cause analysis of defects and suggest solution to implement preventive and corrective measures to address the same.
- Maintains detailed records of activities, discussions and agreed actions in a version control tool (SVN)
- Attends Daily and weekly status Meetings with Client Manager, onsite & offshore team members.
- Creating, renewing, downloading and repairing provisioning profiles.
- Involved in the app market release process for the company.
- Designed screens, implementation of requirements, analysis.
- Issues fixation for the new defects and existed defects.
- Involved in both applications' app store submission (app store & google play).

Environment:

Objective C & Swift, Core Java (android), iOS, Windows, JSON, Core Location, WebKit, Apple Push Notification, COCOA framework, Core Data & SQLite, IOVATION, JUMIO, ACCULYNK, UIAutomation.

Mohawk Corporation, USA **04/2011 – 09/2011**

Senior Software Engineer

Mohawk, is world's leading carpet making company. Mohawk wants to develop mobile applications for their sales executives, this application is in house application. By using this app executives will work to create new carpet or designs based on the end users demand with distributors and company outlets.

Responsibilities:

- Involved in new features implementation.
- Involved in defects fixation.
- Involved in development screens.
- Implemented client-server connections, DB integration.

Environment: XCode, Objective C, JSON, SQL, SQLite.

Signy IT Solutions PVT Ltd, Pune- Developer 08/2010 – 03/2011

iLink iPad Application

iLINK, is an application which contains product details of different companies like Electrolux. In this we can get the details of the company's products, and have a small questionnaire in objective mode about the company product. This is an iPad app which can be used offline to deliver specific knowledge courses to location reps. The app is having the ability to collect data while on the location and later be synced with online system to transfer stored data. (Like personal data, test scores, training information). The application can support a larger system with DB containing reps and all related information

Responsibilities:

- Involved in the implementation of new features.
- Involved in development screens& defect fixes.
- Implemented client-server connections, DB integration.

Environment: Objective C, XCode, JSON parsing, SQLite

Client: Genx, Pune

Quiz game (iPad)

Quiz Game, is an iPad game which contains Quizzes. This game is basically designed & developed to kids (below 5-7 yrs), this can be used in schools too for the education purpose. In this Quizzes are related to basic mathematics, environment, world…etc. This also contains multimedia data like (audio & video), with this kids will get interested to attend quizzes. This quiz can also be customized based on the interest. In this app used good images & animations to interact and to get involve by kids.

Responsibilities:

- Involved in the implementation UI for the game.
- Involved in development screens& defect fixes.

Environment: Objective C, XCode, JSON parsing, SQLite, XML

Client: Todd Albert

Divya Bhaskar & Dainik Bhaskar (News application for iPhone & iPad)

Dainik Bhaskar & Divya Bhaskar are two different applications in English & Gujarati (Indian language). App contains all latest news, city-wise, category-wise…etc. news. In this we used XML parsing to get data from server. We can save the news or we can share the favorite news to friends through twitter, Facebook and through mail.

Responsibilities:

- Involved in the implementation of application both iPhone & iPad.
- Involved in development screens& defect fixes.

Environment: Objective C, COCOA, XCode, XML parsing, Core data, RSS feed

Client: DB Corp Ltd, India

Xerces Technologies PVT Ltd, Pune 07/2009 – 08/2010

Software Developer

Epeler – Funny word game& Bricks (Game for iPhone)

Responsibilities:

- Involved in the design of USX.
- Implemented levels based game.
- Used up to 200K words to validate the words.
- Used peer to peer connection for multi-player.
- Involved in the implementation of collisions using Chipmunk physics engine with Cocos-2d game engine.
- Involved in bug fixes.

Environment: Objective C, COCOA, Cocos-2D, Chipmunk, XCode, SQLite

Client: SkyForce Systems LLC

Simply SUDOKU & Santa Delivery Dash (Game for iPhone)

Responsibilities:

- Involved in the design of USX.
- Implemented levels based game.
- Involved in the implementation of collisions using Chipmunk physics engine with Cocos-2d game engine.
- Involved in bug fixes.
- Implemented 2 level based game.
- Involved in the development with the other team.

Environment: Objective C, COCOA, Cocos-2D, XCode, SQLite

Client: Nine Point Nine Pvt Ltd, Endeavor Brothers Interactive Software

ICE – In Case of Emergency (application for iPhone)

ICE (In Case of Emergency) is an application for iPhone related to health in case of emergency situations. With this user can send his present location (GPS) and saved information for five numbers (SMS) and makes call to first person in the five listed numbers. When the receiver attends the call then loud speaker get activate automatically, so that the person can talk freely. This application is very friendly and easy to use in all emergency situations.

Environment: Objective C, COCOA, XCode, SQLite

Client: Xerces Technologies

Maram Software Solutions, Hyderabad 02/2007 – 05/2009

Software Test Engineer

Pharmacy Management System

The "Pharmacy Management System" aims at providing robust computerized system for maintaining the transactions of medical distributors. The aim of "Pharmacy Management System" is to provide user-friendly interface, which can be easily dealt by people who does not have much knowledge in computers.

The interface is provided with relevant controls and names right from login screen till to the report generation screen. In this application, it's easy to maintain Purchase Orders and Drug Sales, easy to maintain stock details, easy to generate Bills and Reports.

Environment: Manual testing process, Bugzilla, Quality center, MS-Excel

Client: Aurobindo Pharmaceuticals Ltd, Hyderabad.

Customer Relationship Management Application

This is a web based application which powers the sales team of an Organization. It allows the members of sales team to track leads, new Customers, current customers, sales issues and marketing campaigns. It enables sales, marketing and support organizations to manage their Customer interactions more efficiently. The system seamlessly blends all of the functionality required managing information on many aspects of business into a friendly tabbed interface.

The opportunities management in CRM allows measuring sales. Deals by expected close date, amount and incorporates data into pipeline charts and custom reports. The case management allows the user to track customer problems and resolutions

Responsibilities:

- Extensively involved in testing life cycle using black box techniques
- Prepared several Test Cases to the project.
- Expertise in execution of Test Cases.

Environment: Manual testing process, Bugzilla, MS-Excel

Client: In house project.

Web Surveyor

The main objective of the project is to create an instant Survey designer, which can create asp. Net, scripts relative to the requirements. The code generator which will provide the designer code, it can be easily hosted on web sites, and to carryout survey on a global scale. A survey environment with major concentration on authenticity of the person without intervening personal aspects of any person. Invite only a specific group of people to undergo survey. After getting the results the company will calculate the company name and growth in the market

Responsibilities:

- Designed forms that are required for the project.
- Tested all forms with different combinations.

Environment: Manual testing process, MS-Excel

Client: In house project.